UNIT 3 · STANDARD 6.AT.3C

Ratio Reasoning: Convert Measurements Between Systems

Lesson 3-7

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use ratio reasoning to convert measurements between the customary and metric systems.

Language Objective: I can explain a conversion between systems using the words conversion factor, approximately, customary, and metric.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Conversion factor | **Customary system** | **Metric system** | **Approximately** | **Ratio table** | **Reasonableness** | **Multi-step problem**

Level 3: try the blanks from memory first, then check the bank.

1. The ratio that trades one unit for another is a _____ _____.

2. Inches, pounds and gallons belong to the _____ system.

3. Centimeters, grams and liters belong to the _____ system, built on tens.

4. Close to but not exactly equal is _____ — written with ≈.

5. A table whose every column holds the same comparison is a _____ _____.

6. Checking that an answer's size makes sense is checking its _____.

7. A problem that needs two or more operations in order is a _____.

Write About the Math

Is 100 kilometers per hour faster or slower than 100 miles per hour, and how do you know?
¿100 kilómetros por hora es más rápido o más lento que 100 millas por hora, y cómo lo sabes?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Conversion factor** — *Factor de conversión*
- ☐ **Customary system** — *Sistema usual*
- ☐ **Metric system** — *Sistema métrico*
- ☐ **Approximately** — *Aproximadamente*
- ☐ **Ratio table** — *Tabla de razones*

Model: *A mile is longer than a kilometer (1 km ≈ 0.6 mi), so 5 miles is the longer race even though the numbers match.* — Students reason about UNIT SIZE before touching arithmetic — same number, different units, different lengths.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The conversion factor is 1 km ≈ ___ mi. Scaling that up, 100 kilometers ≈ ___ miles, so the sign means about ___ miles per hour. That is ___ than 100 miles per hour, because in one hour you would travel ___ distance.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.